

TWO-AND-A-HALF-DIMENSIONAL FLOWS OF THE H^s EULER–ARNOLD EQUATIONS: A SELF-GRAVITATING BOUSSINESQ SYSTEM

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ABSTRACT. Consider the right-invariant metrics with inertia symbol $a(|k|) = |k|^{2s}$ on $\text{SDiff}(\mathbb{T}^3)$, where $s = 0$ is the incompressible Euler equation. Data independent of one coordinate remain so, but the classical two-and-a-half-dimensional cancellation of the Euler equation survives only at $s = 0$. We show that the reduced dynamics is a Lie–Poisson system on the same semidirect product as the two-dimensional Boussinesq and ideal MHD equations. The swirl momentum θ is transported, and the vertical vorticity q of the momentum is forced by $J(\Lambda^{-2s}\theta, \theta)$: Boussinesq convection in which the gravitational potential is generated by the density itself. Its conserved sup norm is scaling-critical exactly at $s = \frac{1}{2}$, whereas for the Boussinesq system it is supercritical.

We prove the following.

- Global regularity for $s \geq 2$, and a logarithm below it, for inertia $|k|^4/\log(e + |k|)$.
- For $s > \frac{1}{2}$, a continuation criterion in terms of $\nabla\theta$ alone.
- For $1 \leq s < 2$, blowup can occur only through the high–high interaction in a single Jacobian, which transfers fine-scale structure to large-scale vorticity. This is the precise obstruction to regularity in a range that is subcritical for a Casimir and supercritical for the energy.
- No steady state is energy–Casimir stable, and layered states are Jeans-unstable. The obstruction is a problem of unmixing: the source of the vorticity is bounded by a Casimir for all time, and only transport can amplify it. Linear stirring cannot, however strong or time-dependent, so what is needed is folding. The transfer itself is the long-wave Jeans instability of filaments. Its rate decays with the filament thickness exactly when $s > \frac{1}{2}$, and for $s > 1$ it is bounded by a Casimir on every background.

Measurements along solutions, each testing a prediction fixed in advance, leave transport as the only obstruction. The fine filaments never thicken. The one striated bound available grows at a reduced exponential rate, driven by the head of the main arm of the flow. The source is odd under reflections, so symmetry lines act as free-slip walls, and the Hou–Luo stagnation-point geometry is available on the torus without a boundary. We ran a spectral study in that geometry at $s = \frac{1}{4}$, validated to round-off and run at up to 2048^2 grid points. It finds no collapse: the corner of a generic datum filaments with anti-feedback, and the Hou–Luo corner, after a phase of positive feedback that projected blowup, rebounds. This weakens, but does not close, a route to a positive answer to Question 7.1 of Paper I.

1. INTRODUCTION

Fix the bracket of $\text{SDiff}(\mathbb{T}^3)$, and let the inertia operator be the isotropic Fourier multiplier A with symbol $a(|k|) = |k|^{2s}$. The Euler–Arnold equation of this family was studied in [15]. It has a continuation criterion for every $s > 0$, a conserved energy $\|u\|_{H^s}^2$ that is scale-invariant exactly at $s^* = \frac{5}{2}$, and global regularity at and above s^* [13]. Two questions from Paper I organise what is open.

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- Question 7.1 asks whether some isotropic multiplier blows up from smooth data. A positive answer needs one construction. A negative answer contains $A = \text{id}$, and so contains the global regularity of the three-dimensional Euler equation.
- Question 7.2 asks whether completeness holds below s^* , that is, whether the structure of the equation does any work the Sobolev count cannot.

The classical regularity classes of the Euler equation are natural places to test both questions. Two-dimensional flows transfer to every $s > 0$ [15, Prop. 7.1]: the stretching term is absent. This paper concerns the class of *two-and-a-half-dimensional* flows, which are independent of x_3 but have all three components. For the Euler equation it is globally regular by an accident: its stretching term $(w \cdot \nabla)u_3$ equals $\det(\nabla u_3, \nabla u_3)$ and vanishes identically. For $s > 0$ the accident is gone, and something with more structure takes its place.

Results.

- (1) *The reduced system* (Section 2). With $\theta = m_3$ and $q = (\text{curl } m)_3$, the reduced dynamics is

$$\partial_t \theta + u \cdot \nabla \theta = 0, \quad \partial_t q + u \cdot \nabla q = J(\Lambda^{-2s} \theta, \theta), \quad u = \nabla^\perp \psi, \quad q = \Lambda^{2s} \Delta \psi, \quad (1)$$

on $\mathbb{T}^2$, with $J(f, g) = \partial_1 f \partial_2 g - \partial_2 f \partial_1 g$. It is the Lie–Poisson system of the Boussinesq and MHD algebra with a Hamiltonian in which the potential felt by θ is $\Lambda^{-2s} \theta$, the density’s own. This is Boussinesq convection with self-generated gravity, and at $s = 1$ it is the two-dimensional Newtonian potential (Section 3).

- (2) *Criticality* (Section 4). The Casimir $\|\theta\|_{L^\infty}$ is scaling-critical at $s = \frac{1}{2}$ and the energy at $s = 2$. The window $\frac{1}{2} < s < 2$ is subcritical for a conserved quantity of the structure and supercritical for the energy. For the Boussinesq system the same Casimir is supercritical.
- (3) *Theorems* (Section 5).

- Global regularity for $s \geq 2$ (Theorem 5.2), and a logarithm below it, for inertia $|k|^4 / \log(e + |k|)$ (Proposition 5.3).
- For $s > \frac{1}{2}$, breakdown forces $\int \|\nabla \theta\|_{L^\infty} = \infty$ (Theorem 5.4).
- For $1 \leq s < 2$, breakdown forces the high–high part of one Jacobian to become unbounded in $B_{\infty, \infty}^{-2s}$ (Theorem 5.6). That is, blowup requires an inverse transfer from small scales to large ones, which scaling does not see.

- (4) *What the structure gives* (Section 6).

- No steady state is energy–Casimir stable (Theorem 6.1), and layered states are Jeans-unstable (Proposition 6.2).
- The source of q is bounded in $B_{\infty, \infty}^{-2s}$ for all time. Blowup for $s > \frac{1}{2}$ requires transport to amplify it faster than $\log Z$ (Theorem 6.4).
- The inverse transfer is the long-wave Jeans instability of filaments. Its rate decays with the filament thickness exactly when $s > \frac{1}{2}$ (Proposition 6.6), and it is bounded by a Casimir on every background for $s > 1$ (Proposition 6.7).
- Measurements along solutions confirm the picture where it is not proved: transport damps the source on average, and the Jeans rate of the actual filaments falls as they thin.
- What remains is transport alone. The fine filaments never thicken, but the striated bound that would exploit this improves the Lipschitz rate only by a constant factor in the exponent. Its growth comes from the head of the main arm, where the gradient of θ weakens.

- (5) *A numerical study* (Section 7). A symmetry of (1) makes the Hou–Luo stagnation-point geometry available with no boundary. A validated spectral study at $s = \frac{1}{4}$ finds no collapse, and it falsifies a blowup prediction that its own early data supported.

2. THE REDUCTION

As in [15], $m = Au$, and the Euler–Arnold equation is

$$\partial_t m + (u \cdot \nabla)m + (\nabla u)^\top m = -\nabla p, \quad \operatorname{div} u = 0, \quad (2)$$

while $w = \operatorname{curl} m$ obeys the unmodified stretching equation $\partial_t w + (u \cdot \nabla)w = (w \cdot \nabla)u$ for every s [15, Prop. 2.2]. Below, $\Lambda = (-\Delta)^{1/2}$ acts on mean-zero functions on $\mathbb{T}^2$, and $\nabla^\perp = (-\partial_2, \partial_1)$.

Proposition 2.1 (the reduction). *If u_0 is independent of x_3 , so is $u(t)$. Writing $u = (u_h, u_3)$, $\theta = m_3$ and $q = \partial_1 m_2 - \partial_2 m_1$, the pair (θ, q) satisfies (1) with $u = u_h$, and $u_3 = \Lambda^{-2s}\theta$.*

Proof. A commutes with translations, so the x_3 -translate of a solution is a solution. Uniqueness [15, Prop. 4.5] then gives invariance of the class. Take the third component of (2). Nothing depends on x_3 , so $\partial_3 p = 0$ and $((\nabla u)^\top m)_3 = m_j \partial_3 u^j = 0$; hence θ is transported. Take the third component of the stretching equation. There $w_1 = \partial_2 \theta$, $w_2 = -\partial_1 \theta$ and $w_3 = q$, so

$$(w \cdot \nabla)u_3 = \partial_2 \theta \partial_1 u_3 - \partial_1 \theta \partial_2 u_3 = J(u_3, \theta) = J(\Lambda^{-2s}\theta, \theta).$$

Finally $q = \Lambda^{2s}(\partial_1 u_2 - \partial_2 u_1) = \Lambda^{2s}\Delta\psi$ for the stream function ψ of u_h . $\square$

At $s = 0$ the source is $J(\theta, \theta) = 0$, which is the two-and-a-half-dimensional cancellation of the Euler equation. The source is therefore the entire effect of the metric. Each line of (1) agrees with the full three-dimensional right-hand side, evaluated on a grid, to 1.8×10^{-14} at $s = 0, \frac{1}{4}, \frac{1}{2}, 1, 2$ (`derive.py`).

3. STRUCTURE

Proposition 3.1 (conserved quantities). *Along smooth solutions of (1), the following are constant for all functions F and G :*

$$E = \|\Lambda^s u\|_{L^2}^2 + \|\Lambda^{-s} \theta\|_{L^2}^2, \quad \int F(\theta), \quad \int q G(\theta).$$

Proof. $E = \langle m, u \rangle$ is the energy of [15, Lemma 2.5]. $\int F(\theta)$ is constant because θ is transported by a divergence-free field. For the last family, write $P = \Lambda^{-2s}\theta$ and $\tilde{G}' = G$. Then

$$\frac{d}{dt} \int q G(\theta) = \int G(\theta) J(P, \theta) = \int J(P, \tilde{G}(\theta)) = 0,$$

because $J(f, g) = \partial_1(f\partial_2 g) - \partial_2(f\partial_1 g)$ is a divergence. $\square$

Remark 3.2 (the Lie–Poisson form). With $\{f, g\} = J(f, g)$, (1) reads

$$\partial_t \theta + \{\psi, \theta\} = 0, \quad \partial_t q + \{\psi, q\} = \{\delta H / \delta \theta, \theta\}, \quad \psi = -\delta H / \delta q,$$

with Hamiltonian

$$H = \frac{1}{2}E = \frac{1}{2}\langle q, (-\Delta)^{-1}\Lambda^{-2s}q \rangle + \frac{1}{2}\langle \theta, \Lambda^{-2s}\theta \rangle.$$

This is the Lie–Poisson form on the dual of the semidirect product $\mathfrak{sdiff}(\mathbb{T}^2) \ltimes C^\infty(\mathbb{T}^2)$ [9]. The same algebra carries the two-dimensional Boussinesq equations, the ideal MHD equations in flux form, and the two-and-a-half-dimensional Euler equation. The families of Proposition 3.1 are its Casimirs, which is why they do not see the Hamiltonian. The systems differ only through $\delta H / \delta \theta$, and the source is always $J(\delta H / \delta \theta, \theta)$:

system	$\delta H / \delta \theta$	source for q
2D Boussinesq	fixed potential $-x_2$	$\partial_1 \theta$, linear
2D ideal MHD, θ the flux	$-\Delta \theta$	$J(-\Delta \theta, \theta)$, quadratic
(1)	$\Lambda^{-2s}\theta$	$J(\Lambda^{-2s}\theta, \theta)$, quadratic

So (1) is Boussinesq convection in which the gravitational potential is generated by the density itself. At $s = 1$ that potential is the two-dimensional Newtonian potential of θ . The source is quadratic like that of MHD, but smoother by $2 + 2s$ derivatives.

Lemma 3.3 (the Lagrangian identity). *Let Φ solve $\partial_t \Phi + u \cdot \nabla \Phi = \Lambda^{-2s} \theta$ with $\Phi(0) = 0$, and let X_t be the flow of u . Then*

$$q(t) = q_0 \circ X_t^{-1} + J(\Phi(t), \theta(t)).$$

Proof. Write $D_t = \partial_t + u \cdot \nabla$. For divergence-free u in two dimensions, $D_t J(f, g) = J(D_t f, g) + J(f, D_t g)$. The terms containing ∇u sum to $(\operatorname{div} u) J(f, g)$, by the identity $(Ma) \times b + a \times (Mb) = (\operatorname{tr} M)(a \times b)$ for 2×2 matrices M . Hence $D_t(q - J(\Phi, \theta)) = J(\Lambda^{-2s} \theta, \theta) - J(\Lambda^{-2s} \theta, \theta) - J(\Phi, 0) = 0$. $\square$

At $s = 0$ one has $\Phi = t\theta$ and the Jacobian vanishes. The whole departure from the Euler equation is therefore one Jacobian: that of the transported scalar against the time integral, along trajectories, of its own smoothed potential. In a computation against an independent passive scalar (`lagrange.py`), the identity holds to 4×10^{-14} at $s = 1$, where the time step limits the error. It holds to at most 3.2×10^{-8} at $s = \frac{1}{4}$, where the spatial truncation does, while the Jacobian itself is about 0.4 of q .

4. SCALING

Proposition 4.1 (the symmetry). *If (θ, q) solves (1) on $\mathbb{R}^2$, then for all $\lambda, \mu > 0$ so does*

$$\theta_{\lambda, \mu}(x, t) = \mu^{-1} \lambda^{1-2s} \theta(x/\lambda, t/\mu), \quad q_{\lambda, \mu}(x, t) = \mu^{-1} \lambda^{-2s} q(x/\lambda, t/\mu).$$

On $\mathbb{T}^2$ the same holds with $\lambda = 1/n$ for $n \in \mathbb{N}$.

Proof. The operator $\nabla^\perp \Delta^{-1} \Lambda^{-2s}$ is homogeneous of degree $-1 - 2s$, so $u_{\lambda, \mu} = \mu^{-1} \lambda u(x/\lambda, t/\mu)$ and both transport terms balance ∂_t . The source is homogeneous of degree 2 in θ and $2s - 2$ in x , so it scales as $\mu^{-2} \lambda^{2-4s} \lambda^{2s-2}$. That is the scaling of $\partial_t q_{\lambda, \mu}$. $\square$

The symmetry agrees with a grid computation to 9×10^{-14} for $s = \frac{1}{4}, \dots, 1$ (`probe.py`).

Fix the conserved $\|\theta\|_{L^\infty}$. That forces $\mu = \lambda^{1-2s}$: concentrating to scale λ takes time λ^{1-2s} , which tends to zero only when $s < \frac{1}{2}$. So the Casimir is supercritical for $s < \frac{1}{2}$, critical at $s = \frac{1}{2}$, and subcritical for $s > \frac{1}{2}$. Fixing the energy instead forces $\mu = \lambda^{2-s}$, so the energy is critical at $s = 2$. That is the count $s^* = \frac{d}{2} + 1$ of [15, §7] in $d = 2$. For the Boussinesq system, the same computation with the linear source gives $\mu = \lambda^{1/2}$: the sup norm of the density is supercritical there. This is why (1) is not Boussinesq in disguise, even though its naive estimates look the same.

5. THEOREMS

Lemma 5.1 (three kernels on $\mathbb{T}^2$). *Let $s > 0$, and consider operators on mean-zero functions. Then:*

$$(K0) \quad \|\Lambda^{-2s} f\|_{L^\infty} \leq C_s \|f\|_{L^\infty}.$$

$$(K1) \quad \|\nabla \Lambda^{-2s} f\|_{L^\infty} \leq C_s \|f\|_{L^\infty} \text{ if and only if } s > \frac{1}{2}.$$

$$(K2) \quad \text{With } u \text{ recovered from } q \text{ as in (1), } \|\nabla u\|_{L^\infty} \leq C_s \|q\|_{L^\infty}, \text{ with no logarithm.}$$

Proof. The symbols are smooth away from the origin and homogeneous of degrees $-2s$, $1 - 2s$ and $-2s$. Their kernels on $\mathbb{R}^2$ are therefore homogeneous of degrees $-2 + 2s$, $-3 + 2s$ and $-2 + 2s$ [16], and periodisation adds a smooth function. A convolution operator is bounded on L^∞ exactly when its kernel is in L^1 , the norm being $\|K\|_{L^1}$. A homogeneous kernel is locally integrable exactly when its degree exceeds -2 . $\square$

A direct computation of the kernel L^1 norms at $N = 128, \dots, 1024$ (`kernels.py`) shows three behaviours. Above threshold the norms converge. At threshold they grow by exactly 0.441 per doubling of N , which is the logarithm. Below threshold they grow like $2^{1/2-s}$ per doubling, for instance 1.425 against a predicted 1.414 at $s = \frac{1}{4}$ in (K1).

Theorem 5.2 (regularity for $s \geq 2$). *For $s \geq 2$, smooth two-and-a-half-dimensional data on $\mathbb{T}^3$ have global smooth solutions.*

Proof. Put $Z(t) = \int_0^t \|\nabla u\|_{L^\infty(\mathbb{T}^3)} dt$. It suffices to show that Z is finite on bounded intervals [15, Thm. 6.2]. By (K1), $\|\nabla u_3\|_{L^\infty} = \|\nabla \Lambda^{-2s} \theta\|_{L^\infty} \leq C \|\theta_0\|_{L^\infty}$. The energy bounds $\|u_h\|_{H^s(\mathbb{T}^2)}$.

- If $s > 2$, then $H^{s-1}(\mathbb{T}^2) \subset L^\infty$, so $\|\nabla u_h\|_{L^\infty} \leq CE^{1/2}$.
- If $s = 2$, then ∇u_h lies in the critical space $H^1(\mathbb{T}^2)$, and the Brezis–Gallouët inequality [3] gives $\|\nabla u_h\|_{L^\infty} \leq CE^{1/2}(1 + \log(e + \|u\|_{H^r}))^{1/2}$. The momentum-form estimate [15, Lemma 7.3], valid for inertia of order $2s > 1$, gives $\log \|u(t)\|_{H^r} \leq c + CZ(t)$. Hence $Z' \leq C(1 + Z)^{1/2}$, and Z grows at most quadratically.

In either case Z is finite on bounded intervals. $\square$

No structure is used. The effective dimension is two, and the threshold $\frac{5}{2}$ of [13, 15] drops to 2 on this class. As in three dimensions [15, Prop. 7.4], the same count reaches a whole logarithm further. For this proposition only, the inertia symbol is a general $a(|k|)$.

Proposition 5.3 (a logarithm below 2). *Let $a(r) = r^4/g(r)^2$, where $g \geq 1$ is smooth and nondecreasing, $|r^j g^{(j)}(r)| \leq C_j g(r)$, and $g(r) \lesssim r^\varepsilon$ for every $\varepsilon > 0$. Put*

$$H(T) := \int_0^T g(e^\tau)^2 d\tau.$$

If $\int^\infty H(T)^{-1/2} dT = \infty$, then smooth two-and-a-half-dimensional data on $\mathbb{T}^3$ have global smooth solutions. This holds for $a(r) = r^4/\log(e+r)$ and for $a(r) = r^4/(\log(e+r)(\log \log(e^e+r))^2)$. The hypothesis fails for $a(r) = r^4/\log(e+r)^{1+\varepsilon}$.

Proof. The local theory and blowup alternative of [15] use the symbol only through a lower bound $a(r) \gtrsim r^{4-\varepsilon}$, which holds here, as in the proof of [15, Prop. 7.4]. Put $Z(t) = \int_0^t \|\nabla u\|_{L^\infty(\mathbb{T}^3)} dt$. The momentum-form estimate [15, Lemma 7.3] holds for such symbols and gives $\|m(t)\|_{H^\sigma} \leq \|m_0\|_{H^\sigma} e^{CZ(t)}$ for each $\sigma > 0$. So it suffices that Z stay finite on bounded intervals.

The energy $E = \sum_k a(|k|)|\hat{u}(k)|^2$ is conserved, and two-and-a-half-dimensional fields have $\hat{u}(k) = 0$ unless $k_3 = 0$. All sums below therefore run over $k \in \mathbb{Z}^2$, and they cover all three components of u at once. Fix $\rho > 4$. Cauchy–Schwarz against E on $|k| \leq N$, and against $\|u\|_{H^\rho}$ beyond, gives

$$\|\nabla u\|_{L^\infty} \leq \sum_{k \neq 0} |k| |\hat{u}(k)| \leq CE^{1/2} \left(\sum_{0 < |k| \leq N} \frac{g(|k|)^2}{|k|^2} \right)^{1/2} + CN^{2-\rho} \|u\|_{H^\rho}.$$

In two dimensions the lattice sum is at most $C(1 + \int_1^N g(r)^2 dr/r) = C(1 + H(\log N))$. This is the only place where the dimension enters: in [15] the sum runs over $\mathbb{Z}^3$, which is why the symbol there is r^5 . Choose $\log N = \max\{\log 2, \log \|u\|_{H^\rho}/(\rho-2)\}$. Since $\|u\|_{H^\rho} \lesssim \|m\|_{H^{\rho-4+\varepsilon}}$ and $\rho-4+\varepsilon > 0$, this gives $\log N \leq c_0 + CZ(t)$, and so

$$Z'(t) \leq C_1 + C_2 E^{1/2} H(c_1 + c_2 Z(t))^{1/2}.$$

The right side is a positive, continuous, nondecreasing function of Z . By hypothesis its reciprocal is not integrable at infinity, so by Osgood's lemma [2, Lemma 3.4] Z stays finite at every finite time.

For the examples, $g(e^\tau)^2 = \log(e + e^\tau)$ gives $H(T) \sim T^2/2$, and $\int^\infty dT/T = \infty$. The extra factor $(\log \log)^2$ gives $H(T) \sim \frac{1}{2}T^2(\log T)^2$, and $\int^\infty dT/(T \log T) = \infty$. The power $1 + \varepsilon$ gives $H(T) \sim T^{2+\varepsilon}/(2 + \varepsilon)$, and the integral converges. $\square$

At $g \equiv 1$ this is the endpoint case of Theorem 5.2, with $Z' \lesssim 1 + Z^{1/2}$. As in [15, Rem. 7.5], the argument moves no power. What it moves is the baseline in this class. Regularity of two-and-a-half-dimensional flows comes without the structure for every symbol of Proposition 5.3. A result crediting the structure has to get below that whole family. The window $\frac{1}{2} < s < 2$ of Theorems 5.4 and 5.6 lies a full power below it.

Theorem 5.4 (only the transported scalar can break down). *Let $s > \frac{1}{2}$. If the maximal time T_* is finite, then*

$$\int_0^{T_*} \|\nabla \theta(t)\|_{L^\infty} dt = \infty.$$

Proof. Along trajectories, $D_t q = \nabla^\perp(\Lambda^{-2s}\theta) \cdot \nabla \theta$. By (K1) and $\|\theta(t)\|_{L^\infty} = \|\theta_0\|_{L^\infty}$,

$$\|q(t)\|_{L^\infty} \leq \|q_0\|_{L^\infty} + C_s \|\theta_0\|_{L^\infty} \int_0^t \|\nabla \theta\|_{L^\infty} dt.$$

If the integral stayed finite, q would be bounded, so ∇u_h would be bounded by (K2) and ∇u_3 by (K1). Then [15, Thm. 6.2] would continue the solution past T_* . $\square$

Remark 5.5. For $0 < s \leq \frac{1}{2}$ the same argument, with the interpolation $\|\nabla \Lambda^{-2s}\theta\|_{L^\infty} \leq C\|\theta\|_{L^\infty}^{2s-\varepsilon}\|\nabla \theta\|_{L^\infty}^{1-2s+\varepsilon}$, gives $\int_0^{T_*} \|\nabla \theta\|_{L^\infty}^{2-2s+\varepsilon} dt = \infty$ for each $\varepsilon \in (0, 2s)$. Theorem 5.4 is the analogue of the criterion of Chae and Nam [4] and of E and Shu [8] for the Boussinesq equations. In this class it removes q from the criterion of [15] altogether.

Theorem 5.4 leaves open the range $\frac{1}{2} < s < 2$, which is subcritical for the Casimir and supercritical for the energy. The naive estimates there are of Riccati type. From Lemma 3.3 and (K1), (K2), $\|\nabla u\|_{L^\infty} \lesssim \|q_0\|_{L^\infty} + \|\theta_0\|_{L^\infty} \|\nabla \Phi\|_{L^\infty}$, while $\nabla \Phi$ is stretched by ∇u . So $Y = \|\nabla \Phi\|_{L^\infty}$ obeys $Y' \lesssim 1 + Y^2$. Scaling nevertheless forbids self-similar collapse there. The next theorem locates exactly where the estimates lose what scaling promises. Let Δ_j be the Littlewood–Paley blocks, let $S_j = \sum_{i < j} \Delta_i$, and decompose the Jacobian by Bony's paraproduct:

$$J(\Phi, \theta) = \underbrace{\sum_j J(S_{j-1}\Phi, \Delta_j\theta) + \sum_j J(\Delta_j\Phi, S_{j-1}\theta)}_{\text{paraproducts}} + \underbrace{\sum_{|j-j'| \leq 1} J(\Delta_j\Phi, \Delta_{j'}\theta)}_{=: R}.$$

Theorem 5.6 (the obstruction is the inverse transfer). *Let $1 \leq s < 2$. If the maximal time T_* is finite, then*

$$\sup_{t < T_*} \|R(t)\|_{B_{\infty, \infty}^{-2s}} = \infty.$$

Proof. Both paraproducts have spectrum in annuli. By Bernstein's inequality, their block at frequency 2^k is bounded in L^∞ by $C2^{2k}\|\Phi\|_{L^\infty}\|\theta\|_{L^\infty}$. Since $s \geq 1$, their $B_{\infty, \infty}^{-2s}$ norm is at most $C\|\Phi(t)\|_{L^\infty}\|\theta_0\|_{L^\infty}$. By (K0) this is at most $C_s t \|\theta_0\|_{L^\infty}^2$, because Φ is the integral of $\Lambda^{-2s}\theta$ along trajectories. Also $\|q_0 \circ X_t^{-1}\|_{B_{\infty, \infty}^{-2s}} \leq \|q_0\|_{L^\infty}$. So by Lemma 3.3,

$$\|q(t)\|_{B_{\infty, \infty}^{-2s}} \leq \|q_0\|_{L^\infty} + C_s t \|\theta_0\|_{L^\infty}^2 + \|R(t)\|_{B_{\infty, \infty}^{-2s}}.$$

The map $q \mapsto \nabla u_h$ has order $-2s$, so $\|\nabla u_h\|_{B_{\infty,\infty}^0} \leq C\|q\|_{B_{\infty,\infty}^{-2s}}$. The logarithmic inequality of Kozono, Ogawa and Taniuchi [11] gives $\|\nabla u_h\|_{L^\infty} \leq C(1 + \|\nabla u_h\|_{B_{\infty,\infty}^0}(1 + \log^+ \|u\|_{H^r}))$. With [15, Lemma 7.3] as in Theorem 5.2, and (K1) for u_3 ,

$$Z'(t) \leq C(1 + t + \|R(t)\|_{B_{\infty,\infty}^{-2s}})(1 + Z(t)).$$

If $\|R\|_{B_{\infty,\infty}^{-2s}}$ were bounded on $[0, T_*)$, Grönwall's inequality would bound Z there, and [15, Thm. 6.2] would continue the solution. $\square$

Remark 5.7 (reading Theorem 5.6). Four points.

- *Where the damage must land.* Everything in q except R is controlled by conserved quantities and t . R is the part of the Jacobian in which two fields oscillating at the same fine scale 2^{-j} produce output at a coarser scale 2^{-k} , $k \leq j$. Its symbol $\xi \times \zeta$ has size 2^{j+k} , with no further cancellation. The energy controls the coarsest scales, since $\|\Delta_k q\|_{L^\infty} \lesssim 2^{k(2+s)} E^{1/2}$ and hence $2^{-2sk} \|\Delta_k q\|_{L^\infty} \lesssim 2^{k(2-s)} E^{1/2}$. So a blowup must produce this inverse transfer at scales 2^{-k} that themselves tend to zero, while remaining coarse compared with the fields that feed it.
- *The contrast with Boussinesq.* The Boussinesq source $\partial_1 \theta$ is linear and has no high-high part. There the difficulty is plain supercriticality. Here scaling is favourable, and the transfer is what it misses.
- *The Casimirs.* The Casimirs of Proposition 3.1 force the full Jacobian $J(\Phi, \theta) = q - q_0 \circ X^{-1}$ to be orthogonal to every function of θ . That constrains part of the large-scale output that R can produce. By Theorem 6.1 below, they cannot control it.
- *The range $\frac{1}{2} < s < 1$.* The paraproducts also enter there, because $2^{-2sk} 2^{2k}$ is unbounded.

How narrow the obstruction is.

Proposition 5.8 (the same map). *Let $1 \leq s < 2$, let Y be an area-preserving diffeomorphism of $\mathbb{T}^2$, and let φ, ϑ be smooth. For $\Phi = \varphi \circ Y$ and $\theta = \vartheta \circ Y$,*

$$\|R\|_{B_{\infty,\infty}^{-2s}} \leq \|J(\varphi, \vartheta)\|_{L^\infty} + C\|\varphi\|_{L^\infty}\|\vartheta\|_{L^\infty},$$

independently of Y .

Proof. Since $\det \nabla Y = 1$, $J(\varphi \circ Y, \vartheta \circ Y) = J(\varphi, \vartheta) \circ Y$, whose sup norm does not depend on Y . Also $L^\infty \subset B_{\infty,\infty}^{-2s}$. The paraproducts are bounded as in the proof of Theorem 5.6, by $C\|\Phi\|_{L^\infty}\|\theta\|_{L^\infty} = C\|\varphi\|_{L^\infty}\|\vartheta\|_{L^\infty}$. R is the difference. $\square$

Stirring by a single map therefore cannot produce the inverse transfer, however much it stretches. Each gradient grows like the stretching, but the two align, and the Jacobian does not grow. In the dynamics $\theta(t) = \theta_0 \circ X_t^{-1}$ has exactly this form. Φ does not, and the obstruction lies in the time dependence of the potential it accumulates. R is bilinear, $\theta(t) = \theta(\tau) \circ X_{t \rightarrow \tau}$ and $\Phi(t) = \int_0^t P(\tau) \circ X_{t \rightarrow \tau} d\tau$, where $X_{t \rightarrow \tau}$ carries positions at time t back to time τ and $P = \Lambda^{-2s} \theta$. Hence

$$R(t) = \int_0^t R(P(\tau) \circ X_{t \rightarrow \tau}, \theta(\tau) \circ X_{t \rightarrow \tau}) d\tau. \quad (3)$$

This is a superposition of same-map pushes of the pairs $(P(\tau), \theta(\tau))$. Proposition 5.8 bounds each term by $\|J(P(\tau), \theta(\tau))\|_{L^\infty}$, which is large, because the base pair is *incoherent*: P is smooth and $\theta(\tau)$ is mixed. That only returns Theorem 5.4.

Incoherent pairs are, however, harmless in $B_{\infty,\infty}^{-2s}$ too. We checked this with exact iterates Y^n of an area-preserving standard-shear map, sampled at the grid points without interpolation, at $N = 1024$ (`obstruction.py`).

- *Same map.* For $\Phi = \varphi \circ Y^n$ and $\theta = \vartheta \circ Y^n$, $\|J\|_{L^\infty} = 1.4500$ at every n . Meanwhile $\|R\|_{B_{\infty,\infty}^{-2s}}$ stays between 0.77 and 0.91 while $\|\nabla\theta\|_{L^\infty}$ grows 49-fold.
- *Independent maps.* With Φ stirred by a different map family, $\|J\|_{L^\infty}$ grows 3500-fold, yet $\|R\|_{B_{\infty,\infty}^{-2s}}$ stays between 0.9 and 2.2. The high-high output of incoherent fine-scale fields lands at high frequencies, where the weight 2^{-2sk} suppresses it.

What Theorem 5.6 needs for a blowup is therefore narrow: fine-scale co-oscillation of Φ and θ that is coherent but phase-shifted. It is neither the aligned case, which the flow maps produce from a fixed pair, nor the independent one. By Remark 5.7, it must moreover land at scales that drift finer, since the energy caps every fixed coarse scale.

In the dynamics. We evolved Φ alongside the solution and measured R directly (`sympy.py`), for the bump datum of Section 7. That datum has $q_0 = 0$, so $q = J(\Phi, \theta)$ exactly; this identity, and the exactness of the Bony split, serve as checks. At $s = 1$:

- $\|R\|_{B_{\infty,\infty}^{-2s}}$ rises from 0.002 to about 0.5 by $t \approx 49$, and then levels off through $t = 60$.
- Over the same time $\|\nabla\Phi\|_{L^\infty}$ grows 3000-fold and $\|\nabla\theta\|_{L^\infty}$ 13-fold.
- $M = 256$ and $M = 512$ agree to five digits through $t = 44$.

The peak of the scale profile $2^{-2sk}\|\Delta_k R\|_{L^\infty}$ sits at the coarsest block. The intermediate blocks grow at rates of 0.1 to 0.2 per unit time, which is the migration toward finer scales described in Remark 5.7. They stay below the coarse peak: at $t = 44$ the profile over $k = 0, \dots, 5$ is (0.42, 0.31, 0.052, 0.081, 0.016, 0.001). At $M = 512$ the migration continues to $t = 60$. The coarse blocks level off near 0.45 and 0.39, while blocks $k = 2, 4, 5$ keep growing, to 0.14, 0.13 and 0.017, and none reaches the coarse peak. The last frame is marginal: the check $q = J(\Phi, \theta)$ degrades there to 3×10^{-2} , against 2×10^{-4} at $t = 54$. The fine-to-coarse output of the finest blocks runs at a nearly constant 4×10^{-4} of the size those blocks would produce with no cancellation. At $s = 1.5$ the picture is the same: $\|R\|_{B_{\infty,\infty}^{-2s}}$ grows linearly to 0.38 at $t = 60$. The obstruction is thus present in principle, suppressed in practice, and not bounded by any argument given so far. The transfer observed here is dipolar and runs along level lines, so it is orthogonal to every function of θ , and the Casimir orthogonality $\int J(\Phi, \theta)G(\theta) = 0$ does not see it. Section 6 takes the obstruction apart.

6. WHAT THE STRUCTURE GIVES, AND WHERE THE OBSTRUCTION LIVES

This section takes the obstruction of Theorem 5.6 apart, in four steps.

- (1) The Casimirs give no stability at all (Theorem 6.1).
- (2) The obstruction is a problem of unmixing. The source of q is bounded by a Casimir for all time, and blowup needs transport to amplify it faster than a logarithm (Proposition 6.3, Theorem 6.4). Linear stirring never does, so the amplification has to come from folding (Proposition 6.5).
- (3) The inverse transfer is the Jeans instability of filaments, and a Casimir bounds its gain (Propositions 6.6, 6.7 and Remark 6.8).
- (4) Measurements along solutions locate what is left, which is transport.

Numerical statements carry the name of their script. Where a statement tests a prediction, the prediction was fixed before the run.

Arnold's method is empty. Let H be the Hamiltonian of Remark 3.2, and for functions F, G put

$$H_C = H + \int F(\theta) + \int q G(\theta).$$

Since $\delta H/\delta q = -\psi$ and $\delta H/\delta \theta = \Lambda^{-2s}\theta$, a pair $(\bar{\theta}, \bar{q})$ is a critical point of H_C exactly when

$$G(\bar{\theta}) = \bar{\psi}, \quad \Lambda^{-2s}\bar{\theta} + F'(\bar{\theta}) + \bar{q}G'(\bar{\theta}) = 0. \quad (4)$$

Such a pair is a steady state. Arnold's method [1, 10] gives its nonlinear stability whenever the second variation of H_C there is definite.

Theorem 6.1 (no energy-Casimir stability). *Let $s > 0$, let $F, G \in C^2$, and let $(\bar{\theta}, \bar{q})$ be smooth, satisfy (4), and have $\bar{\theta}$ non-constant. Then the second variation of H_C at $(\bar{\theta}, \bar{q})$ is indefinite.*

Proof. The second variation is

$$\delta^2 H_C = \langle \delta q, (-\Delta)^{-1} \Lambda^{-2s} \delta q \rangle + \langle \delta \theta, \Lambda^{-2s} \delta \theta \rangle + \int (F''(\bar{\theta}) + \bar{q}G''(\bar{\theta})) \delta \theta^2 + 2 \int G'(\bar{\theta}) \delta q \delta \theta.$$

It is positive at $\delta \theta = 0$, so we must rule out $\delta^2 H_C \geq 0$. Suppose it holds. Take a cutoff χ , numbers a, b , and $\delta \theta = b\chi \cos(Nx_1)$, $\delta q = a\chi \cos(Nx_1)$. As $N \rightarrow \infty$ these tend weakly to zero. The two nonlocal operators are compact, since their symbols decay, so the first two terms tend to zero. The local terms tend to $\frac{1}{2} \int \chi^2 (2abG'(\bar{\theta}) + b^2(F''(\bar{\theta}) + \bar{q}G''(\bar{\theta})))$. Nonnegativity for all a and all χ forces $G'(\bar{\theta}(x)) = 0$ at every x .

The range $I = \bar{\theta}(\mathbb{T}^2)$ is a non-degenerate interval. So G is constant on I , and $G'' = 0$ there. By (4), $\bar{\psi}$ is constant, so $\bar{q} = 0$. The same limit now gives $F'' \geq 0$ on I . Hence $\Psi = -F'$ is non-increasing on I , and $\Lambda^{-2s}\bar{\theta} = \Psi(\bar{\theta})$. Since $\theta = \Lambda^{2s}u_3$ has mean zero, $0 \in I$, and

$$\|\Lambda^{-s}\bar{\theta}\|_{L^2}^2 = \int \bar{\theta} \Psi(\bar{\theta}) = \int \bar{\theta} (\Psi(\bar{\theta}) - \Psi(0)) \leq 0,$$

because the two factors of the integrand have opposite signs. So $\bar{\theta} = 0$, a contradiction. $\square$

The proof uses only two facts. The Casimirs are local and linear in q , and the energy is compact at small scales. Adding linear conserved functionals, such as momenta, does not change the second variation, so the localisation step survives them. Theorem 6.1 is the functional-analytic form of the observation that closes Section 5: the Casimirs see nothing of q at small scales.

The first steady states to test are the layered ones, and they are unstable.

Proposition 6.2 (Jeans instability of layered states). *Let $s > 0$, and let $\bar{\theta} = \bar{\theta}(x_2)$ be smooth, non-constant and of mean zero, with $\Lambda^{-2s}\bar{\theta} = \Psi(\bar{\theta})$ for some $\Psi \in C^1$. This holds, for instance, for $\bar{\theta} = f(\cos nx_2)$ with $f' \neq 0$. Then $(\bar{\theta}, 0)$ is a steady state, and the linearised system has solutions growing like $e^{\sigma t}$ with $\sigma > 0$.*

Proof. Every Jacobian of two functions of x_2 vanishes, and $u = 0$, so the state is steady. Put $B = \bar{\theta}'(x_2) \partial_1$, $L = (-\Delta)^{-1} \Lambda^{-2s}$, $r = \Psi'(\bar{\theta})$ and $K = r - \Lambda^{-2s}$. Using $\psi' = -Lq'$ and $\partial_2 \Lambda^{-2s}\bar{\theta} = r\bar{\theta}'$, the linearisation is

$$\partial_t \theta' = BLq', \quad \partial_t q' = -BK\theta', \quad \text{so} \quad \partial_t^2 \theta' = \mathcal{M}K\theta', \quad \mathcal{M} = B^*LB = -BLB.$$

$\mathcal{M}$ is non-negative and compact, of order $-2s$. The nonzero spectrum of $\mathcal{M}K$ equals that of the compact self-adjoint operator $\mathcal{M}^{1/2}K\mathcal{M}^{1/2}$. If $\lambda > 0$ is an eigenvalue of the latter, with eigenfunction w , then $v = \mathcal{M}^{1/2}w$ satisfies $\mathcal{M}Kv = \lambda v$. Then $\theta' = e^{\sqrt{\lambda}t}v$ and $q' = -\lambda^{-1/2}e^{\sqrt{\lambda}t}BKv$ solve the linearisation. The range of $\mathcal{M}^{1/2}$ is dense in $(\ker \mathcal{M})^\perp$, and $\ker \mathcal{M} = \ker B$ because L is positive. It therefore suffices to find $v \perp \ker B$ with $\langle Kv, v \rangle > 0$.

By the last step of the proof of Theorem 6.1, Ψ is not non-increasing on $\bar{\theta}(\mathbb{T})$. So $r > 0$ on an open set U on which $\bar{\theta}' \neq 0$. Functions in $\ker B$ are independent of x_1 on U , so any v supported in U with zero x_1 -mean is orthogonal to $\ker B$. For $v = \chi \cos(Nx_1)$ with χ supported in U , $\langle Kv, v \rangle \rightarrow \frac{1}{2} \int r\chi^2 > 0$ as $N \rightarrow \infty$. $\square$

For $\bar{\theta} = \cos x_2$ we have $r \equiv 1$ and $K = 1 - \Lambda^{-2s} \geq 0$, so every x_1 -wavenumber k_1 is unstable. At $s = 0$, $K = 0$, and the layered states of the two-and-a-half-dimensional Euler equation are neutral, as they must be. A truncated eigenvalue computation (`jeans.py`) converges to ten digits and gives maximal rates $\sigma = 0.452, 0.411, 0.505$ and 0.571 at $s = \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}$. It is checked against the nonlinear solver of Section 7, started at $\cos x_2$ plus a perturbation of size 10^{-8} . In the symmetric class at $k_1 = 1$, the measured rates are 0.20122 and 0.22827 , against predicted rates of 0.20136 and 0.22824 , at $s = \frac{1}{4}$ and 1 . The stratification is unstable at every $s > 0$, as a self-gravitating one should be. No layered state can serve as a stable reference, and no steady state is energy–Casimir stable.

The obstruction is unmixing.

Proposition 6.3 (the source is bounded; only transport can amplify it). *Let $s > \frac{1}{2}$, $P = \Lambda^{-2s}\theta$ and $W = \nabla^\perp \theta$.*

- (i) $J(P, \theta) = \operatorname{div}(\theta \nabla^\perp P)$, so $\|J(P(t), \theta(t))\|_{B_{\infty, \infty}^{-2s}} \leq C_s \|\theta_0\|_{L^\infty}^2$ for all t .
- (ii) $q(t) = q_0 \circ X_t^{-1} + \int_0^t J(P(\tau), \theta(\tau)) \circ X_{t \rightarrow \tau} d\tau$.
- (iii) W is frozen in, $\partial_W \theta = 0$ and $[D_t, \partial_W] = 0$. Moreover $J(\Phi, \theta) = -\partial_W \Phi$, and the source is a commutator: $D_t q = -\partial_W P = -[\partial_W, \Lambda^{-2s}]\theta$.
- (iv) $\|J(P(\tau), \theta(\tau)) \circ X_{t \rightarrow \tau}\|_{B_{\infty, \infty}^{-2s}} \leq C_s \|\theta_0\|_{L^\infty}^2 e^{Z(t) - Z(\tau)}$.

Proof. (i) One has $J(f, g) = \operatorname{div}(g \nabla^\perp f)$. For $s \geq \frac{1}{2}$, $\|\operatorname{div} F\|_{B_{\infty, \infty}^{-2s}} \leq C \|F\|_{L^\infty}$, and $\|\nabla P\|_{L^\infty} \leq C_s \|\theta\|_{L^\infty}$ by (K1). (ii) This is Duhamel's formula for $D_t q = J(P, \theta)$. (iii) θ is transported and u is divergence-free, so $\nabla^\perp \theta$ is transported as a vector field, and derivatives along transported fields commute with D_t . Also $\partial_W \theta = \nabla^\perp \theta \cdot \nabla \theta = 0$ and $\partial_W f = -J(f, \theta)$. Hence $\partial_W P = [\partial_W, \Lambda^{-2s}]\theta + \Lambda^{-2s} \partial_W \theta = [\partial_W, \Lambda^{-2s}]\theta$. (iv) By the same-map identity, $J(P, \theta) \circ Y = \operatorname{div}((\theta \circ Y) \nabla^\perp (P \circ Y))$ for area-preserving Y , and $\|\nabla(P \circ X_{t \rightarrow \tau})\|_{L^\infty} \leq \|\nabla P\|_{L^\infty} e^{Z(t) - Z(\tau)}$. $\square$

So the source of q is bounded in $B_{\infty, \infty}^{-2s}$ for all time, by a Casimir. The size of q in that norm is created entirely by the transport in (ii). Transport enlarges a negative norm only by *unmixing*, that is, by moving fine-scale structure to coarser scales. By (iii), what is transported is a commutator: the failure of Λ^{-2s} to commute with differentiation along the level lines of θ . The continuation argument needs only a logarithm.

Theorem 6.4 (the obstruction, sharpened). *Let $s > \frac{1}{2}$ and suppose the maximal time T_* is finite. Then*

$$\sup_{t < T_*} \frac{\|q(t)\|_{B_{\infty, \infty}^{-2s}}}{\log(e + Z(t))} = \infty.$$

For $1 \leq s < 2$ the same holds with R in place of q .

Proof. The proof of Theorem 5.6 used $s \geq 1$ only to bound the paraproducts. Without them it gives $Z' \leq C(1 + \|q\|_{B_{\infty, \infty}^{-2s}})(1 + Z)$ for $s > \frac{1}{2}$. If $\|q\|_{B_{\infty, \infty}^{-2s}} \leq K \log(e + Z)$ on $[0, T_*)$, then $Z' \leq C(1 + K)(1 + Z) \log(e + Z)$. Since $\int_0^\infty dz / ((1 + z) \log(e + z)) = \infty$, Osgood's lemma keeps Z finite at T_* , and [15, Thm. 6.2] continues the solution. For $1 \leq s < 2$, the proof of Theorem 5.6 gives $\|q\|_{B_{\infty, \infty}^{-2s}} \leq \|q_0\|_{L^\infty} + C_s t \|\theta_0\|_{L^\infty}^2 + \|R\|_{B_{\infty, \infty}^{-2s}}$. $\square$

Define the *unmixing ratio*

$$\operatorname{Mix}(t) = \frac{\|q(t) - q_0 \circ X_t^{-1}\|_{B_{\infty, \infty}^{-2s}}}{\int_0^t \|J(P(\tau), \theta(\tau))\|_{B_{\infty, \infty}^{-2s}} d\tau}.$$

It is the time-averaged factor by which transport has amplified the source, in the norm that matters. By Proposition 6.3, $\|q(t)\|_{B_{\infty,\infty}^{-2s}} \leq \|q_0\|_{L^\infty} + C_s t \|\theta_0\|_{L^\infty}^2 \text{Mix}(t)$. The Lipschitz bound (iv) gives $\text{Mix} \lesssim e^Z$, and with it Theorem 6.4 yields only the Riccati inequality $Z' \lesssim t e^Z (1 + Z)$. Regularity requires $\text{Mix} \lesssim \log(e + Z)$. That is an exponential improvement on the Lipschitz bound, and it is exactly what a mixing estimate would have to supply.

No such estimate can hold for arbitrary bounded sources. A flow that mixes forward in time unmixes whatever it is handed already mixed. Near a hyperbolic stagnation point, for example, a source oscillating along the expanding direction is coarsened at the rate of the strain. An estimate must therefore use the particular source in (ii), which is built from $\theta(\tau)$, mixed already by the same flow over $[0, \tau]$. The Casimirs are linear in q , so by Theorem 6.1 they cannot supply it.

The particular source does help, and against every linear stirring it suffices. Near a hyperbolic stagnation point of a strong background flow, and for small θ , the flow is to leading order linear, $u = G(t)x$ with $\text{tr } G = 0$, on $\mathbb{R}^2$. Let F_t be its flow map, so $F'_t = G F_t$, $F_0 = I$ and $\det F_t = 1$. In this kinematic model θ is transported, $P = a(D)\theta$ for a symbol with $0 \leq a \leq 1$, and $D_t \Phi = P$ with $\Phi(0) = 0$. The symbol $a(k) = (1 + |k|^2)^{-s}$ carries the infrared cutoff of $\mathbb{T}^2$.

Proposition 6.5 (linear stirring cannot unmix the source). *Let $s > 0$, and let θ_0 be smooth with $\nabla \theta_0$ in the Wiener algebra $\mathcal{A}$, the space of functions whose Fourier transform is integrable. For every linear flow, however it depends on time,*

$$J(\Phi(t), \theta(t)) = J(m_t(D)\theta_0, \theta_0) \circ F_t^{-1}, \quad m_t(K) = \int_0^t a(F_\tau^{-\top} K) d\tau \in [0, t],$$

and hence

$$\|J(\Phi(t), \theta(t))\|_{B_{\infty,\infty}^{-2s}} \leq C_s \|J(\Phi(t), \theta(t))\|_{L^\infty} \leq C_s t \|\nabla \theta_0\|_{\mathcal{A}} \|\nabla \theta_0\|_{L^\infty}.$$

The bound does not depend on F , so it holds however large $Z = \int_0^t |G|$ is.

Proof. Since $\theta(t) = \theta_0 \circ F_t^{-1}$, we have $\hat{\theta}(t, k) = \hat{\theta}_0(F_t^\top k)$. Put $\tilde{\Phi} = \Phi(t) \circ F_t$. Then $\partial_t \tilde{\Phi} = P(t) \circ F_t$, whose Fourier transform in the Lagrangian variable is $a(F_t^{-\top} K) \hat{\theta}(t, F_t^{-\top} K) = a(F_t^{-\top} K) \hat{\theta}_0(K)$. Integrating in time gives $\hat{\tilde{\Phi}} = m_t \hat{\theta}_0$. For $\det F = 1$, $J(f \circ F^{-1}, g \circ F^{-1}) = J(f, g) \circ F^{-1}$, which gives the identity. Then $\|\nabla \tilde{\Phi}\|_{L^\infty} \leq \|m_t |K| \hat{\theta}_0\|_{L^1} \leq t \|\nabla \theta_0\|_{\mathcal{A}}$, and $|J(f, g)| \leq |\nabla f| |\nabla g|$. Composition with F_t^{-1} does not change sup norms, and $\|f\|_{B_{\infty,\infty}^{-2s}} \leq C_s \|f\|_{L^\infty}$ for $s > 0$. $\square$

So the numerator of Mix grows at most linearly in t , whatever the strain does. This extends Proposition 5.8 from a single map to the accumulated potential of a time-dependent map, and the reason is the same. $P(\tau) \circ F_\tau$ is a Fourier multiplier applied to θ_0 , so the potential stays coherent with the scalar it came from.

The proof also shows what a nonlinear flow must do to break this. For any area-preserving flow map X_τ and any kernel k_a of $a(D)$,

$$\nabla_\xi (P(\tau) \circ X_\tau)(\xi) = \int k_a(X_\tau(\xi) - y) [\nabla X_\tau^{-1}(y) \nabla X_\tau(\xi)]^\top \nabla \theta_0(X_\tau^{-1}(y)) dy. \quad (5)$$

For linear maps the bracket is the identity. In general it compares the deformation gradients at two points a unit Eulerian distance apart, and for a folding flow it can be of size e^{2Z} . The obstruction of Theorem 6.4 therefore lives in the variation of ∇X across the scale on which $a(D)$ averages, that is, in folding. Straining, however strong and however it rotates, is harmless. A direct computation for steady, alternating and rotating strains (`kinematic.py`) agrees, and can do no more: the proposition covers every linear flow. A numerical test of folding hits the

resolution limit at $Z \approx 6$, whether the fine scales are taken in the Eulerian or the Lagrangian variable.

Filaments: the inverse transfer is a Jeans instability. A mixed θ consists of filaments, which are locally layered. Consider a layer of thickness $1/n$, $\bar{\theta} = A \cos nx_2$, displaced by a velocity that varies along it at wavenumber k_1 . The displacement $-u' \cdot \nabla \bar{\theta}$ is proportional to $\bar{\theta}'$: a copy of the layer shifted by a quarter period. That is exactly the coherent, phase-shifted pair singled out after Proposition 5.8 as the dangerous kind. Its Jacobian with the layer lands at wavenumber k_1 . For $k_1 = 1$ this is the coarsest mode, since the vorticity of the mode of Proposition 6.2 with $\varphi = e^{ix_1}$ is $q' = L^{-1}\varphi = e^{ix_1}$. The linear Jeans loop $q' \rightarrow u' \rightarrow \theta' \rightarrow q'$ about a filament is therefore the inverse transfer of Theorem 5.6, and its growth rate is the rate of that transfer.

Put $\varphi = Lq'$. The linearisation of Proposition 6.2 becomes $\partial_t^2 \varphi = LB^*KB\varphi$, so the largest growth rate σ satisfies

$$\sigma^2 = \sup_{\varphi} \frac{\langle KB\varphi, B\varphi \rangle}{\|\Lambda^{1+s}\varphi\|_{L^2}^2}. \quad (6)$$

Proposition 6.6 (the scale law). *Let $s > 0$, and let $\sigma(n)$ be the largest growth rate about $\bar{\theta} = A \cos nx_2$. Then*

$$\max\left\{n^{1-2s}\sigma(1), A\left(\frac{1}{2}n^2(n^{-2s} - (1+n^2)^{-s})\right)^{1/2}\right\} \leq \sigma(n) \leq C_s A n^{\max(1-2s, -s)}.$$

In particular $\sup_n \sigma(n) < \infty$ if and only if $s \geq \frac{1}{2}$. The second lower bound, attained by the long-wave mode $k_1 = 1$, is $A\sqrt{s/2}n^{-s}(1 + O(n^{-2}))$.

Proof. Lower bounds. The $2\pi/n$ -periodic functions form an invariant subspace, and Proposition 4.1 with $\lambda = 1/n$ and $\mu = n^{2s-1}$ maps the problem at $n = 1$ onto it at the same amplitude, multiplying rates by n^{1-2s} . For the second bound, insert $\varphi = e^{ix_1}$ in (6). Then $B\varphi$ sits at $(1, \pm n)$, where $K = n^{-2s} - (1 + n^2)^{-s}$, and $\|B\varphi\|^2 = A^2 n^2/2$.

Upper bound. Fix $k_1 \neq 0$, since $k_1 = 0$ gives $B\varphi = 0$. Write $\varphi = e^{ik_1 x_1} \sum_p f_p e^{ipx_2}$, $D_p = k_1^2 + p^2$ and $\kappa(m) = n^{-2s} - (k_1^2 + m^2)^{-s}$. Then

$$\langle KB\varphi, B\varphi \rangle = \frac{1}{4} k_1^2 A^2 n^2 \sum_m \kappa(m) |f_{m-n} - f_{m+n}|^2, \quad \|\Lambda^{1+s}\varphi\|^2 = \sum_p D_p^{1+s} |f_p|^2.$$

It suffices to bound the numerator by $\sum_p \alpha_p D_p^{1+s} |f_p|^2$ with $\alpha_p \leq C_s A^2 \max(n^{-2s}, n^{2-4s})$.

- If $k_1 \geq n/2$, use $\kappa \leq n^{-2s}$ and $\sum_m |f_{m-n} - f_{m+n}|^2 \leq 4 \sum_p |f_p|^2$. This gives $\alpha_p \leq A^2 n^{2-2s} k_1^{-2s} \leq 4^s A^2 n^{2-4s}$.
- If $k_1 < n/2$, split $f = f^L + f^H$ at $|p| = n/2$.
 - *Low part.* Each m meets at most one p with $|p| < n/2$, and convexity of $x \mapsto x^{-s}$ gives $\kappa(p+n) + \kappa(p-n) \leq 2(n^{-2s} - (n^2 + D_p)^{-s}) \leq 2sn^{-2s-2}D_p$. So $\alpha_p \leq \frac{s}{2} A^2 n^{-2s} D_p^{1-s}$ with $1 \leq D_p \leq n^2/2$.
 - *High part.* Use $\kappa \leq n^{-2s}$ and $D_p \geq n^2/4$, which gives $\alpha_p \leq 4^s A^2 n^{2-4s}$.
 - *Cross terms.* They pair $|p| < n/2$ with $p \pm 2n$, where $|\kappa| \leq 4^s n^{-2s}$ and $D_{p \pm 2n} \geq n^2$. Cauchy-Schwarz with weights D^{1+s} gives $\alpha \leq 4^s A^2 n^{1-3s} D_p^{(1-s)/2}$.

In each case $\alpha \leq C_s A^2 \max(n^{-2s}, n^{2-4s})$. $\square$

The eigenvalue computation (`jeans.py`) matches both lower bounds. At $n = 4, 8, 16$ and 32 , the long-wave rate agrees with the formula to four digits for $s = \frac{1}{4}, \dots, \frac{3}{2}$. The maximal rate follows n^{1-2s} exactly for $s \leq \frac{1}{2}$, and n^{-s} for $s \geq 1$. At $s = \frac{1}{4}$ it is 0.43, 0.86 and 1.71 at $n = 1, 4, 16$. At $s = \frac{1}{2}$ it is 0.411 at every n . At $s = 1$ it is 0.51, 0.17 and 0.044.

So a layer of thickness $1/n$ transfers to the coarsest scale at rate $\approx A\sqrt{s/2}n^{-s}$, and more slowly the thinner it is. Replacing k_1 by $m \ll n$ gives the transfer from scale $1/n$ to scale $1/m$ at rate $\approx A\sqrt{s/2}m^{1-s}n^{-s}$, which is at most $An^{\max(1-2s, -s)}$. For $s \geq \frac{1}{2}$ no filament, however thin, transfers faster than a Casimir allows. For $s < \frac{1}{2}$ thin filaments are the fastest. The threshold is again the Casimir-critical exponent of Section 4.

For $s > 1$ the uniform bound does not need layers.

Proposition 6.7 (the Jeans loop is Casimir-bounded). *Let $s > 1$, and let $\bar{\theta}$ be any smooth background, possibly depending on t , with $\bar{P} = \Lambda^{-2s}\bar{\theta}$. Consider the linearisation about $\bar{\theta}$ with the advection by the background velocity removed:*

$$\partial_t \theta' = J(Lq', \bar{\theta}), \quad \partial_t q' = J(\Lambda^{-2s}\theta', \bar{\theta}) + J(\bar{P}, \theta').$$

Then $E_2 = \frac{1}{2}\langle q', Lq' \rangle + \frac{1}{2}\langle \theta', \Lambda^{-2s}\theta' \rangle$ satisfies $E_2(t) \leq E_2(0) \exp(C_s \int_0^t \|\bar{\theta}\|_{L^\infty})$.

Proof. Write $Bf = J(f, \bar{\theta})$. B is antisymmetric, because $\nabla^\perp \bar{\theta}$ is divergence-free. Hence

$$\dot{E}_2 = \langle Lq', B\Lambda^{-2s}\theta' \rangle + \langle Lq', J(\bar{P}, \theta') \rangle + \langle \Lambda^{-2s}\theta', BLq' \rangle = \langle Lq', V \cdot \nabla \theta' \rangle, \quad V = \nabla^\perp \bar{P}.$$

Integrating by parts, $|\dot{E}_2| \leq \|V \cdot \nabla Lq'\|_{H^s} \|\theta'\|_{H^{-s}}$. Now V has order $1 - 2s$ in $\bar{\theta}$, so $\|V\|_{B_{\infty, \infty}^{2s-1}} \leq C_s \|\bar{\theta}\|_{L^\infty}$. Since $2s - 1 > s$, multiplication by V is bounded on H^s . Also $\|\nabla Lq'\|_{H^s}^2 \leq \langle q', Lq' \rangle$. So $|\dot{E}_2| \leq C_s \|\bar{\theta}\|_{L^\infty} E_2$. $\square$

The coupling to the background enters E_2 only through the smooth field $\nabla^\perp \bar{P}$; the filaments' gradients, which drive the loop, cancel in the energy identity.

On the actual filaments. Propositions 6.6 and 6.7 cover layers, and every background for $s > 1$. To test the picture where it is not proved, we froze snapshots $\theta(t)$ of the bump solution of Section 7. We unfolded them to the full torus, so that no symmetry is imposed on the perturbation. We then computed the top of the spectrum of the frozen operator, $\partial_t^2 \theta' = J(L[J(\Lambda^{-2s}\theta', \bar{\theta}) + J(\bar{P}, \theta')], \bar{\theta})$, with ARPACK at $N = 512$ (`jeans_frozen.py`). On layered backgrounds this reproduces `jeans.py`. The prediction, fixed before the runs, was this: for $s \geq \frac{1}{2}$ the rate $\sigma(t)$ stays of the order of its initial value while θ filaments, and at $s = \frac{1}{4}$ it grows roughly like $\|\nabla \theta\|_{L^\infty}^{1/2}$.

s	resolved to	$\ \nabla \theta\ _{L^\infty}$	σ
0.25	$t = 12$	$2.0 \rightarrow 9.0$	$0.44 \rightarrow 0.82$
0.75	$t = 28$	$2.0 \rightarrow 12.3$	$0.120 \rightarrow 0.086$
1	$t = 44$	$2.0 \rightarrow 14.9$	$0.085 \rightarrow 0.051$
1.5	$t = 56$	$2.0 \rightarrow 8.8$	$0.055 \rightarrow 0.048$

In the window, the rate falls as the filaments thin. At $s = \frac{1}{4}$ it rises, with effective exponent 0.41 overall and 0.47 over the last interval. At the end of the run at $s = 1$ the frozen rate is a third of $\|\nabla u\|_{L^\infty} = 0.15$.

Remark 6.8 (the norm gap is an artifact of linearising). Proposition 6.7 is an energy estimate, $2 - s$ derivatives weaker than $B_{\infty, \infty}^{-2s}$, and it does not lift with Casimir-only constants. About $A \cos nx_2$ at $k_1 = 1$, the largest amplification beyond the rate grows like $n^{0.99}$ in the energy norm. In the norm lifted by $r = 2 - s + 0.1$, the level that controls $\nabla u'$, it grows like $n^{2.09}$ at $s = 1$ and $n^{1.60}$ at $s = 1.5$. The ratios $n^{1.10}$ and $n^{0.61}$ are n^r , as predicted before the computation (`jeans_lift.py`). The linearisation lets a coarse vorticity perturbation displace a filament without limit, far past its own thickness, which gives the factor n ; lifting multiplies it by the fineness raised to the gap. The nonlinear problem does not need the lift. There the

displacement is capped by the conserved $\|\theta\|_{L^\infty}$, and the self-gravity reaches $\|q\|_{B_{\infty,\infty}^{-2s}}$ only through the source, which Proposition 6.3(i) bounds by a Casimir in this very norm.

In the linear theory, the only producer of coherent, phase-shifted pairs is the Jeans loop, and on filaments its gain is bounded by a Casimir. In the nonlinear problem the self-gravity enters $B_{\infty,\infty}^{-2s}$ only through a Casimir-bounded source. So the obstruction does not lie in the self-gravity. It is transport alone: the question left by Proposition 6.3 is whether the flow can unmix.

Transport, measured. Three measurements along the bump solutions of Section 7 locate what the transport does.

The unmixing ratio. $\text{Mix}(t)$ is computable. We added it to the runs of Section 5, at $M = 256$ (`symcoh.py`), together with a direct measure of coherence. For each block j , let J_j be the high-high Jacobian of $\Delta_j \Phi$ against $\Delta_{j-1} \theta + \Delta_j \theta + \Delta_{j+1} \theta$, and let κ_j be the fraction of its L^2 norm that lands at $|k| < 2^{j-2}$, at least four times coarser. On wave packets at $j = 5$ the measure gives 0.88 for a phase-shifted pair, 0.09 for an in-phase pair, and 10^{-17} for independent wavevectors.

- $s = 1$, resolved to $t = 44$. Mix falls from 1 to 0.83 at $t \approx 30$ and ends at 0.89. Over the same run e^Z reaches 66.
- $s = 0.75$, resolved to $t = 29$. Mix falls to 0.84 and ends at 0.87, with $e^Z = 62$.
- $s = 1.5$, resolved to $t = 83$. Mix falls to 0.81 at $t \approx 50$. Between $t \approx 53$ and 71, $\|q\|_{B_{\infty,\infty}^{-2s}}$ grows faster than the source, and Mix reaches 1.006 at $t = 72$. It returns to 0.93 by $t = 83$, while e^Z goes from 7 to 99.

At $s = 1.5$, an independent run at $M = 512$ agrees with $M = 256$ through $t = 60$. Mix agrees to 5×10^{-5} and $\|q\|_{B_{\infty,\infty}^{-2s}}$ to five digits. Once the blocks are populated, κ_5 and κ_6 agree to three digits, and the check $q = J(\Phi, \theta)$ holds to 4×10^{-12} . The last frames at $s = 1.5$ are marginal: that check degrades from 3.5×10^{-4} at $t = 72$ to 8×10^{-3} at $t = 83$.

Transport has therefore damped the source on average, not amplified it. The episode at $s = 1.5$ is not an inverse transfer from fine scales. Its growth sits in the coarsest block, $k = 0$ ($0.43 \rightarrow 0.59$), which the energy caps (Remark 5.7). The energy is conserved to all digits shown. The kinetic energy, fed by the potential energy $\|\Lambda^{-s} \theta\|^2$ as θ mixes, is near its maximum during the episode. What grows is its concentration in that block: $\|\Delta_0 q\|_{L^\infty}$ rises from 0.49 of the largest value the current kinetic energy permits at $t = 52$ to 0.64 at $t = 72$ (`symenergy.py`). Mix is thus blunter than it looks: it records every growth of $\|q\|_{B_{\infty,\infty}^{-2s}}$ beyond the source, including redistribution among the energy-capped coarse blocks. Coherence, however, does build. In the resolved fine blocks $j = 5, 6, 7$, κ_j rises to 0.3–0.45, between the in-phase and the phase-shifted calibrations, and it rises as the filaments thin. Coherent, phase-shifted fine-scale pairs are present, but their coarse output stays small.

Strain geometry. A filament of θ , and with it the oscillation of the source, thickens only where the strain stretches along its normal. Along trajectories $D_t \log |\nabla \theta| = -\lambda_n$, where $\lambda_n = n \cdot S n = \gamma \cos 2\alpha$ and $n = \nabla \theta / |\nabla \theta|$. Here γ is the strain rate and α the angle between n and the stretching direction. Filaments are made with their level lines along the stretching direction, so unmixing them needs the strain to turn through a right angle against the level lines. This is a geometric condition in the spirit of Constantin, Fefferman and Majda [5], and we measured it (`symstrain.py`), weighting by $|\nabla \theta_{\text{fine}}|^2$ with θ_{fine} the part of θ at $|k| \gtrsim 16$.

- The weighted mean of λ_n is negative at every output time of every run.
- After an initial transient, the weighted mean of $\cos 2\alpha$ sits between -0.96 and -0.98 . The level lines stay within 6° to 8° of the stretching direction.
- The θ -variance flux into every band $|k| \gtrsim 2^j$, $j = 3, \dots, 6$, is positive throughout.

We had registered a prediction that the episode at $s = 1.5$ would coincide with the strain turning against the level lines. It failed on every criterion, with a correlation of -0.04 , for the reason found above: that episode is coarse and energy-capped, not an unmixing of filaments. The fine filaments do not thicken in these runs.

The striated remainder. To see whether this geometry can be made into a bound, resolve $\nabla\Phi$ against the level lines of θ at time t : $\nabla\Phi = \lambda\nabla\theta + \beta\nabla^\perp\theta$. Then $\lambda = \partial_n\Phi/|\nabla\theta|$ and $\beta|\nabla\theta| = \partial_\tau\Phi = -(q - q_0 \circ X_t^{-1})/|\nabla\theta|$. Since $J(\Phi, \theta) = \operatorname{div}((\theta - c)\nabla^\perp\Phi)$ for every constant c , Lemma 3.3 gives exactly, in the sense of distributions for Morse θ ,

$$q - q_0 \circ X_t^{-1} = -(\theta - c)J(\lambda, \theta) - \operatorname{div}(\beta(\theta - c)\nabla\theta). \quad (7)$$

Put $T_1(c) = \|(\theta - c)\partial_\tau\Phi\|_{L^\infty}$ and $T_2(c) = \|(\theta - c)J(\lambda, \theta)\|_{L^{3/2}}$. At critical points of θ , $J(\lambda, \theta)$ has an integrable singularity of order $1/r$, and $L^{3/2} \subset B_{\infty,\infty}^{-4/3} \subset B_{\infty,\infty}^{-2s}$ for $s \geq \frac{2}{3}$. So $\|q - q_0 \circ X_t^{-1}\|_{B_{\infty,\infty}^{-2s}} \leq C(T_1(c) + T_2(c))$ for every c .

The first term is controlled by thickening. W is frozen in, so $D_t(\partial_\tau\Phi) = \partial_\tau P + \lambda_n \partial_\tau\Phi$, and thinning damps $\partial_\tau\Phi$. Along trajectories,

$$|\partial_\tau\Phi(t)| \leq \|\nabla P\|_{L^\infty} \int_0^t \frac{|\nabla\theta(\tau)|}{|\nabla\theta(t)|} d\tau,$$

where the ratio is taken along the trajectory.

We measured both terms (`syntau.py`, `symlocal.py`). While resolved, the curl identity behind (7) holds to between 10^{-13} and 2×10^{-5} . The prediction, registered before the runs, was that both terms grow at most at half the exponential rate of $\|\nabla\Phi\|_{L^\infty}$. The constant matters for where the terms peak. The bump datum leaves 72% of the torus on a plateau where θ equals its far-field value c_* to within e^{-22} ; there $|\nabla\theta| \approx 0$ and $q = 0$. With $c = 0$, the supremum of $|\partial_\tau\Phi|$ moves late onto that plateau, an artifact of the constant. With $c = c_*$ the plateau drops out, but the rates do not change. Here G denotes the logarithmic growth over the resolved window.

s	window	$G_{\nabla\Phi}$	$G_{T_1(c_*)}/G_{\nabla\Phi}$	$G_{T_2(c_*)}/G_{\nabla\Phi}$
0.75	$t = 10 \rightarrow 20$	1.25	0.53	0.44
1	$t = 10 \rightarrow 32$	1.86	0.60	0.57
1.5	$t = 10 \rightarrow 64$	2.64	0.70	0.69

Including the marginal final frames, the ratios are 0.61–0.79. The terms never grow logarithmically.

The supremum of $T_1(c_*)$ lies inside the bump material, where $\theta - c_* \approx 0.5$ – 0.7 , near the crest of the main arm along the stretching axis. It travels outward with the arm's head: x_1 goes from about 0.5 to 2.0, 2.5 and 2.8 at $s = 0.75$, 1 and 1.5. There $|\nabla\theta|$ falls from 0.6–0.7 of its maximum to 0.04–0.1 of it. By the bound above, the gradient must weaken along trajectories there. The head of the arm thickens, although the fine filaments do not.

The striated form of the transport bound thus improves the Lipschitz rate by a constant factor in the exponent, not by the exponential that Theorem 6.4 requires. A conditional theorem built on it would assume what the solutions do not do, and we have not built one.

Summary. The self-gravity is bounded by a Casimir wherever the regularity problem needs it. What is left is transport. Averaged over the domain, transport damps the source, and the fine filaments never thicken. The only striated bound available still grows exponentially, driven by the thickening head of the main arm. The mixing estimate of Theorem 6.4 remains the missing piece, and the head of the arm is where it would have to be won.

7. A TARGETED SEARCH FOR COLLAPSE AT $s < \frac{1}{2}$

Question 7.1 of Paper I needs one example. By Section 4, the class (1) at $s < \frac{1}{2}$ is the cheapest place to look: it is two-dimensional, the Casimir is supercritical, and the following symmetry supplies the geometry of the known Boussinesq blowups without a boundary.

Proposition 7.1 (free walls). *The class of θ even in x_1 and in x_2 , and q odd in both, is invariant under (1). On it:*

- (1) $u_2 = 0$ on $\{x_2 = 0\}$ and $u_1 = 0$ on $\{x_1 = 0\}$, so the origin is a hyperbolic stagnation point;
- (2) the source vanishes on both axes;
- (3) on $\{x_2 = 0\}$, with $P = \Lambda^{-2s}\theta$,

$$\partial_2 J(P, \theta) = \partial_1 P \partial_{22} \theta - \partial_{22} P \partial_1 \theta.$$

Proof. A Jacobian changes sign under each reflection. With θ and P even, the source is therefore odd in each variable, like q . The remaining claims are parity bookkeeping. $\square$

This geometry is not available for the Boussinesq system. There $\partial_1 \theta$ is even under $x_2 \mapsto -x_2$, which is why the proofs of Chen and Hou [6, 7] require a genuine boundary. Here the axes are free-slip walls. The price is (2): the forcing on a wall appears one order up, as the Boussinesq term $\partial_1 \theta$ with the variable coefficient $-\partial_{22} P$. The invariance, the walls and the vanishing hold to 2×10^{-13} , and the forcing identity to 1.6×10^{-12} (`symmetry.py`).

Method. The symmetric class is expanded in cosine series for θ and sine series for q on the quarter domain. With $M + 1$ points per direction this is the full-torus resolution $N = 2M$ at a quarter of the cost.

- *Scheme.* Type-I discrete cosine and sine transforms, the 2/3 dealiasing rule, and RK4 with Courant number 0.4.
- *Stopping rule.* A fixed spectral-tail criterion: the fraction of $\|\nabla \theta\|_{L^2}^2$ or $\|q\|_{L^2}^2$ in the top fifth of the retained modes reaching 10^{-4} .
- *Validation.* Before any production run, the solver was compared with an independent full-torus FFT solver on identical data. The two agree to 4×10^{-16} in θ and 7×10^{-16} in q after 100 steps.
- *Conservation.* Across all runs the energy is conserved to 5×10^{-11} , and $\int \theta^4$ to 1.5×10^{-6} at the stopping time.

Two data were fixed in advance, with $q_0 = 0$ and the mean of θ removed:

- a generic elliptical bump, $\theta_0 = \exp((\cos x_1 - 1)/0.36 + (\cos x_2 - 1)/0.09)$;
- the Hou–Luo configuration [12], $\theta_0 = \sin^2 x_1 \exp((\cos x_2 - 1)/0.09)$, which vanishes at the stagnation point and peaks along the axis.

Since the source is quadratic, $\theta \mapsto -\theta$ gives the same flow. We ran $s \in \{0.1, 0.25, 0.4\}$, with $s = 0.75$ as the subcritical control, at $M = 256$ and 512 , and at $M = 1024$ for the bump at $s = \frac{1}{4}$.

Diagnostics, and a failure mode. For a self-similar singularity of (1), both $1/\|\nabla u\|_{L^\infty}$ and $\|\nabla \theta\|_{L^\infty}^{-(1-2s)}$ fall linearly to zero at the same time T_* . Two estimates that agree look like strong evidence. They are not.

- At $M = 256$ for the bump they agreed (16.5 against 16.0 at $s = 0.1$) over a window in which $\|\nabla \theta\|_{L^\infty}$ grew only threefold, while the growth was visibly *decelerating*.
- $\|\nabla u\|_{L^\infty}$ can also migrate away from the collapse point, into the shear along a filament.

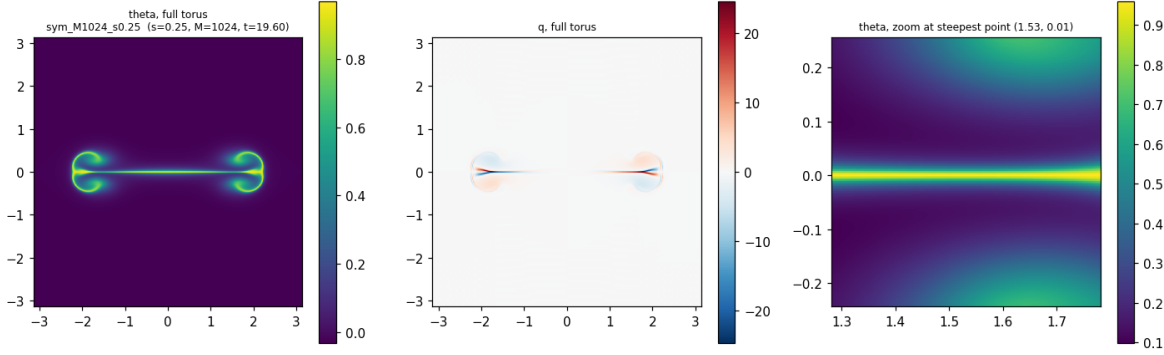


FIGURE 1. The bump at $s = \frac{1}{4}$, $M = 1024$, $t = 19.6$. Left: θ . Centre: q . Right: θ near the steepest gradient, the neck behind a mushroom head. Fields are unfolded to the full torus by the symmetries.

The discriminating diagnostics are two. The first is the trend of the growth rate $d \ln \|\nabla \theta\|_{L^\infty} / dt$, which must diverge. The second is the feedback exponent at a corner, $d \ln |a| / d \ln \kappa$, where $a = \partial_1 u_1$ at the corner and κ is the curvature of θ along the compressed axis. A self-similar collapse needs $|a| \sim \kappa^{(1-2s)/2}$.

The bump. The flow builds a stagnation point at the origin and squeezes the bump into a filament, a dumbbell whose heads become mushroom caps carrying vortex dipoles (Figure 1).

- *Anti-feedback at the corner, for every s .* At $M = 512$ the strain at the origin falls as the filament thins, like ℓ^p with $p = +0.72, +0.89, +0.96, +1.20$ for $s = 0.1, 0.25, 0.4, 0.75$, against the $\ell^{-(1-2s)}$ that collapse requires. At $s = \frac{1}{4}$ it falls from its maximum 0.24 to 0.055.
- *A real but slow late intensification.* Later, the steepest gradient moves along the filament to the neck behind each head. This is converged:

t	$M = 256$	$M = 512$	$M = 1024$
10	6.246	6.248	6.251
12	9.014	9.038	9.077
15	—	15.96	15.96
16	—	19.65	19.97

(values of $\|\nabla \theta\|_{L^\infty}$). But it is not a collapse:

- the growth rate per unit time is 0.16, 0.19, 0.21, 0.22, 0.24, 0.25, 0.24 over $12 \leq t \leq 19$;
- its reciprocal levels off near 4 instead of falling to zero;
- the two estimates of T_* disagree (20.9 against 25.6);
- the neck is still 5 to 11 grid cells wide when the run stops at $t = 19.6$, with $\|\nabla \theta\|_{L^\infty}$ grown 25-fold.

This is the picture E and Shu found for the Boussinesq bubble cap [8]: an intensifying front whose strain saturates.

The Hou–Luo configuration. This datum has two corners.

- *The θ -maximum corner $(\frac{\pi}{2}, 0)$.* It behaves like the bump: a sheet thins at a decelerating rate, $0.32 \rightarrow 0.18$.

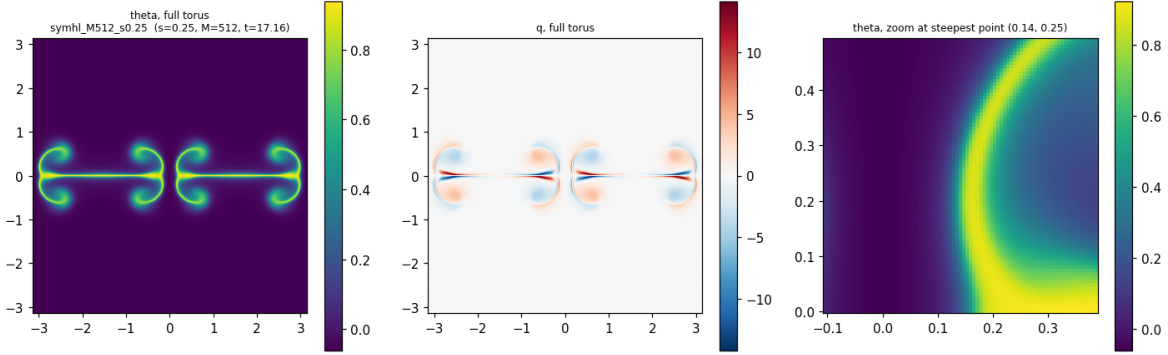


FIGURE 2. The Hou–Luo configuration at $s = \frac{1}{4}$, $M = 512$, $t = 17.2$, after the rebound. Left: θ . Centre: q . Right: θ near the steepest gradient, on a cap turning away from the corner.

- *The Hou–Luo corner* at the origin, where $\theta = 0$. The flow squeezes the profile of θ along the axis toward it, and this showed genuine positive feedback. At $M = 256$, $|a| \sim \kappa^{0.90}$, above the 0.25 that collapse needs. Integrating that law from the last resolved state projected blowup near $t \approx 19$.

The prediction fails at $M = 512$, in a window converged to about 1% against $M = 256$:

t	10.0	11.8	13.6	14.5	15.4	16.3	17.2
κ	3.90	5.16	7.13	8.58	10.38	12.57	11.27
a	-0.075	-0.085	-0.097	-0.105	-0.097	-0.049	+0.050

The compression peaks at $t \approx 14.5$ and then reverses. The dipoles approaching the corner rebound and curl off along the extensional axis (Figure 2), as head-on dipoles do in two-dimensional Euler flow.

Controls. At $s = 0.75$ both data grow exponentially: rate 0.071 for the bump, with no finite-time estimate, and decelerating corners throughout. In the Hou–Luo control the corner compression weakens from 0.080 to 0.039. The diagnostics therefore separate the regimes, as scaling requires.

What this shows. Transplanted to the symmetric setting, the Hou–Luo mechanism does not ignite at $s = \frac{1}{4}$. A generic stagnation point filaments with anti-feedback. What survives is the intensification of a plume neck, smooth, resolved, and at most slowly accelerating. The study does not exclude blowup: it uses two data and finite resolution. Deciding whether the neck follows an exponential with a slowly rising rate or a very slow power law would need the adaptive meshes of [12], and a proof would need the machinery of [6, 7].

8. DISCUSSION

For Question 7.1 of Paper I, the class (1) remains the cheapest arena for a positive answer, since blowup of two-and-a-half-dimensional data is blowup of smooth data on $\mathbb{T}^3$. But the two scenarios that the geometry makes available both limit themselves, and for reasons visible in the fields. Nothing here bears on a negative answer, which contains the Euler equation.

For Question 7.2, Theorems 5.2, 5.6 and 6.4 bracket the problem inside this class. Regularity holds without the structure for $s \geq 2$, and a logarithm below (Proposition 5.3). For $1 \leq s < 2$ it fails only if the high–high part of one Jacobian transfers fine-scale structure to coarser scales

faster than $\log Z$. For $s > \frac{1}{2}$ it fails only if transport amplifies a source that a Casimir bounds for all time. The Casimirs $\int q G(\theta)$ constrain part of that transfer, but by Theorem 6.1 they cannot control it.

Open question. *Is (1) globally regular for some s with $\frac{1}{2} < s < 2$? Equivalently, for $1 \leq s < 2$: does the high-high part of $J(\Phi, \theta)$ stay bounded in $B_{\infty, \infty}^{-2s}$ along solutions?*

An affirmative answer for any $s < 2$ would be the first instance, in this programme, of the Lie structure doing work that the Sobolev count cannot, in a class where vortex stretching is present. The measurements suggest the answer is yes for $s > \frac{1}{2}$. In every resolved window the quantity of Theorem 5.6 stays of order one, and transport damps the source of q on average instead of amplifying it.

Section 6 narrows what is missing to transport.

- The Casimirs cannot supply the estimate (Theorem 6.1).
- The self-gravity needs no further estimate. The producer of coherent, phase-shifted pairs is the Jeans loop of filaments, whose gain a Casimir bounds. In the nonlinear problem the self-gravity enters $B_{\infty, \infty}^{-2s}$ only through a Casimir-bounded source, and the norm gap of the linear theory is an artifact of linearising.
- In these runs the fine filaments never thicken. But the only striated bound we have grows at a reduced exponential rate. The growth comes from the head of the main arm, where the gradient of θ weakens along trajectories.

A mixing estimate of the kind Theorem 6.4 needs would have to control that head.

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The scripts underlying the computational statements are available from the author. They are the following: `derive.py`, `lagrange.py`, `probe.py`, `kernels.py`, `symmetry.py`, `symsolve.py`, `symcheck.py`, `symrun.py`, `symanalyze.py`, `corners.py`, `symfig.py`, `obstruction.py`, `symphi.py`, `jeans.py`, `symcoh.py`, `jeans_frozen.py`, `symstrain.py`, `symenergy.py`, `syntau.py`, `symlocal.py`, `jeans_lift.py` and `kinematic.py`.

REFERENCES

- [1] V. I. Arnold, *On an a priori estimate in the theory of hydrodynamical stability*, Izv. Vyssh. Uchebn. Zaved. Mat. (1966), no. 5, 3–5; English transl. Amer. Math. Soc. Transl. (2) **79** (1969), 267–269.
- [2] H. Bahouri, J.-Y. Chemin and R. Danchin, *Fourier Analysis and Nonlinear Partial Differential Equations*, Grundlehren der mathematischen Wissenschaften **343**, Springer, 2011.
- [3] H. Brezis and T. Gallouët, *Nonlinear Schrödinger evolution equations*, Nonlinear Anal. **4** (1980), 677–681.
- [4] D. Chae and H.-S. Nam, *Local existence and blow-up criterion for the Boussinesq equations*, Proc. Roy. Soc. Edinburgh Sect. A **127** (1997), 935–946.
- [5] P. Constantin, C. Fefferman and A. J. Majda, *Geometric constraints on potentially singular solutions for the 3-D Euler equations*, Comm. Partial Differential Equations **21** (1996), 559–571.

- [6] J. Chen and T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis*, [arXiv:2210.07191](#).
- [7] J. Chen and T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data II: Rigorous numerics*, [arXiv:2305.05660](#).
- [8] W. E and C.-W. Shu, *Small-scale structures in Boussinesq convection*, Phys. Fluids **6** (1994), 49–58.
- [9] D. D. Holm, J. E. Marsden and T. S. Ratiu, *The Euler–Poincaré equations and semidirect products with applications to continuum theories*, Adv. Math. **137** (1998), 1–81.
- [10] D. D. Holm, J. E. Marsden, T. Ratiu and A. Weinstein, *Nonlinear stability of fluid and plasma equilibria*, Phys. Rep. **123** (1985), 1–116.
- [11] H. Kozono, T. Ogawa and Y. Taniuchi, *The critical Sobolev inequalities in Besov spaces and regularity criterion to some semi-linear evolution equations*, Math. Z. **242** (2002), 251–278.
- [12] G. Luo and T. Y. Hou, *Potentially singular solutions of the 3D axisymmetric Euler equations*, Proc. Natl. Acad. Sci. USA **111** (2014), 12968–12973.
- [13] E. Olson and E. S. Titi, *Viscosity versus vorticity stretching: global well-posedness for a family of Navier–Stokes-alpha-like models*, Nonlinear Anal. **66** (2007), no. 11, 2427–2458.
- [14] A. Salkimbayev, *Rigidity of the Euler–Arnold form under energy-preserving deformation*, companion paper.
- [15] A. Salkimbayev, *A Beale–Kato–Majda criterion for right-invariant H^s metrics on $\text{SDiff}(\mathbb{T}^3)$* , companion paper.
- [16] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, 1970.

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